

Is Mrs. Gallas a Good Free Throw Shooter?



Mrs. Gallas was a star basketball player in high school, where she was an impressive 70% free throw shooter.

1. In your own words, what does it mean to be a 70% free throw shooter?

Each free throw has a 70% chance of going in.

2. If Mrs. Gallas takes 100 shots, how many do you expect she will make? Is this guaranteed?

70, No

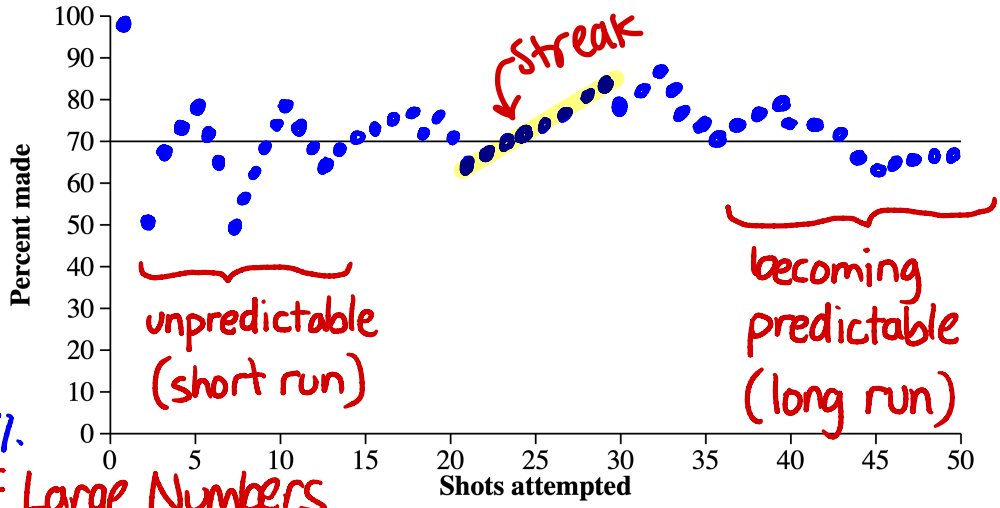
3. If Mrs. Gallas takes 100 shots, do you think she will have a streak where she makes 10 in a row?

No streaks are actually common!

4. Go to stapplet.com/largenum.html. Use the applet to shoot 100 free throws and use the table to keep track of the results.

Shot #	1	2	3	4	5	10	15	20	30	40	50	100
Result (make or miss)	✓	×	✓	✓	✓	✓	✓	✓	✓	×	✓	✓
Percent made	100	50	67	75	80	70	72	70	74	72	68	69

5. Sketch the graph displaying the percent made after each shot attempted.



6. What do you notice in the graph?

lots of variability at first and then tending towards 70%.

← Law of Large Numbers

7. In the 100 shots, did Mrs. Gallas make exactly 70 free throws? Why or why not?

No, every random sample of 100 shots is different.

← sampling variability

8. Based on the results of the applet, what does a probability of 70% mean?

After many, many free throws, the percent of makes will approach 70%.

→ long run relative frequency

Lesson 3.1 – Introducing Probability

QuickNotes

LT*1 Probability

Long run relative frequency

- always between 0 & 1 (inclusive)
- short run is unpredictable
- long run is predictable

LT*2 Law of Large Numbers

Simulated probabilities tend to get closer to the true probability as the number of trials increases.

Check Your Understanding

1. Choose a probability value to complete the middle column of the table, then give an example of an event that may have this probability. Probabilities: 0, 0.001, 0.3, 0.6, 0.99, 1

Explanation	Probability	Example
This outcome is impossible. It can never occur.	0	Rolling a 7 on a fair die
This outcome is certain. It will occur on every trial.	1	Rolling a 1-6 on a fair die
This outcome is very unlikely, but it will occur once in a while in a long sequence of trials.	0.001	Rockford beats East Kentwood at a track meet
This outcome will occur more often than not, but doesn't occur almost every time.	0.6	Sunny day in Kentwood

2. In the Wheel of Fortune, there is a 1/9 = 11.1% probability of spinning "Bankrupt" on any given spin. Interpret the probability.

After many, many spins on the Wheel of Fortune, the percent of times spinning "Bankrupt" approaches 11.1% in the long run.

3. In 9 consecutive spins of the Wheel of Fortune, none of the spins land on "Bankrupt". The next contestant worries that he is more likely to land on Bankrupt now because Bankrupt is due. Explain why this thinking is wrong.

Probability is unpredictable in the short run and streaks are common.